



# A



## BIP39 word sheet A of 4 · words 1–512 of 2,048, in official order

Each die is a coin. Read the top face only: ● **dots**, ○ **blank** (ordinary dice: even ●, odd ○). Every word is equally likely. The first four letters (**bold**) identify every word uniquely.

**Dice:** sheet **2 orange** | row **5 white** | column **4 black** · within each colour, fewest dots on the left

| row      | ○○○○<br>1       | ○○○●<br>2       | ○○●○<br>3      | ○○●●<br>4       | ○○○○<br>5      | ○○●●<br>6       | ○○○○<br>7       | ○○●●<br>8       | ●○○○<br>9       | ●○○●<br>10      | ●○○○<br>11      | ●○○●<br>12      | ●●○○<br>13      | ●●○○<br>14      | ●●○○<br>15      | ●●●●<br>16     |
|----------|-----------------|-----------------|----------------|-----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|
| ○○○○○ 1  | <b>abandon</b>  | <b>ability</b>  | <b>able</b>    | <b>about</b>    | <b>above</b>   | <b>absent</b>   | <b>absorb</b>   | <b>abstract</b> | <b>absurd</b>   | <b>abuse</b>    | <b>access</b>   | <b>accident</b> | <b>account</b>  | <b>accuse</b>   | <b>achieve</b>  | <b>acid</b>    |
| ○○○○● 2  | <b>acoustic</b> | <b>acquire</b>  | <b>across</b>  | <b>act</b>      | <b>action</b>  | <b>actor</b>    | <b>actress</b>  | <b>actual</b>   | <b>adapt</b>    | <b>add</b>      | <b>addict</b>   | <b>address</b>  | <b>adjust</b>   | <b>admit</b>    | <b>adult</b>    | <b>advance</b> |
| ○○○○○ 3  | <b>advice</b>   | <b>aerobic</b>  | <b>affair</b>  | <b>afford</b>   | <b>afraid</b>  | <b>again</b>    | <b>age</b>      | <b>agent</b>    | <b>agree</b>    | <b>ahead</b>    | <b>aim</b>      | <b>air</b>      | <b>airport</b>  | <b>aisle</b>    | <b>alarm</b>    | <b>album</b>   |
| ○○○○● 4  | <b>alcohol</b>  | <b>alert</b>    | <b>alien</b>   | <b>all</b>      | <b>alley</b>   | <b>allow</b>    | <b>almost</b>   | <b>alone</b>    | <b>alpha</b>    | <b>already</b>  | <b>also</b>     | <b>alter</b>    | <b>always</b>   | <b>amateur</b>  | <b>amazing</b>  | <b>among</b>   |
| ○○○○○ 5  | <b>amount</b>   | <b>amused</b>   | <b>analyst</b> | <b>anchor</b>   | <b>ancient</b> | <b>anger</b>    | <b>angle</b>    | <b>angry</b>    | <b>animal</b>   | <b>ankle</b>    | <b>announce</b> | <b>annual</b>   | <b>another</b>  | <b>answer</b>   | <b>antenna</b>  | <b>antique</b> |
| ○○○○● 6  | <b>anxiety</b>  | <b>any</b>      | <b>apart</b>   | <b>apology</b>  | <b>appear</b>  | <b>apple</b>    | <b>approve</b>  | <b>april</b>    | <b>arch</b>     | <b>arctic</b>   | <b>area</b>     | <b>arena</b>    | <b>argue</b>    | <b>arm</b>      | <b>armed</b>    | <b>armor</b>   |
| ○○○○○ 7  | <b>army</b>     | <b>around</b>   | <b>arrange</b> | <b>arrest</b>   | <b>arrive</b>  | <b>arrow</b>    | <b>art</b>      | <b>artefact</b> | <b>artist</b>   | <b>artwork</b>  | <b>ask</b>      | <b>aspect</b>   | <b>assault</b>  | <b>asset</b>    | <b>assist</b>   | <b>assume</b>  |
| ○○○○● 8  | <b>asthma</b>   | <b>athlete</b>  | <b>atom</b>    | <b>attack</b>   | <b>attend</b>  | <b>attitude</b> | <b>attract</b>  | <b>auction</b>  | <b>audit</b>    | <b>august</b>   | <b>aunt</b>     | <b>author</b>   | <b>auto</b>     | <b>autumn</b>   | <b>average</b>  | <b>avocado</b> |
| ○○○○○ 9  | <b>avoid</b>    | <b>awake</b>    | <b>aware</b>   | <b>away</b>     | <b>awesome</b> | <b>awful</b>    | <b>awkward</b>  | <b>axis</b>     | <b>baby</b>     | <b>bachelor</b> | <b>bacon</b>    | <b>badge</b>    | <b>bag</b>      | <b>balance</b>  | <b>balcony</b>  | <b>ball</b>    |
| ○○○○● 10 | <b>bamboo</b>   | <b>banana</b>   | <b>banner</b>  | <b>bar</b>      | <b>barely</b>  | <b>bargain</b>  | <b>barrel</b>   | <b>base</b>     | <b>basic</b>    | <b>basket</b>   | <b>battle</b>   | <b>beach</b>    | <b>bean</b>     | <b>beauty</b>   | <b>because</b>  | <b>become</b>  |
| ○○○○○ 11 | <b>beef</b>     | <b>before</b>   | <b>begin</b>   | <b>behave</b>   | <b>behind</b>  | <b>believe</b>  | <b>below</b>    | <b>belt</b>     | <b>bench</b>    | <b>benefit</b>  | <b>best</b>     | <b>betray</b>   | <b>better</b>   | <b>between</b>  | <b>beyond</b>   | <b>bicycle</b> |
| ○○○○● 12 | <b>bid</b>      | <b>bike</b>     | <b>bind</b>    | <b>biology</b>  | <b>bird</b>    | <b>birth</b>    | <b>bitter</b>   | <b>black</b>    | <b>blade</b>    | <b>blame</b>    | <b>blanket</b>  | <b>blast</b>    | <b>bleak</b>    | <b>bless</b>    | <b>blind</b>    | <b>blood</b>   |
| ○○○○○ 13 | <b>blossom</b>  | <b>blouse</b>   | <b>blue</b>    | <b>blur</b>     | <b>blush</b>   | <b>board</b>    | <b>boat</b>     | <b>body</b>     | <b>boil</b>     | <b>bomb</b>     | <b>bone</b>     | <b>bonus</b>    | <b>book</b>     | <b>boost</b>    | <b>border</b>   | <b>boring</b>  |
| ○○○○● 14 | <b>borrow</b>   | <b>boss</b>     | <b>bottom</b>  | <b>bounce</b>   | <b>box</b>     | <b>boy</b>      | <b>bracket</b>  | <b>brain</b>    | <b>brand</b>    | <b>brass</b>    | <b>brave</b>    | <b>bread</b>    | <b>breeze</b>   | <b>brick</b>    | <b>bridge</b>   | <b>brief</b>   |
| ○○○○○ 15 | <b>bright</b>   | <b>bring</b>    | <b>brisk</b>   | <b>broccoli</b> | <b>broken</b>  | <b>bronze</b>   | <b>broom</b>    | <b>brother</b>  | <b>brown</b>    | <b>brush</b>    | <b>bubble</b>   | <b>buddy</b>    | <b>budget</b>   | <b>buffalo</b>  | <b>build</b>    | <b>bulb</b>    |
| ○○○○● 16 | <b>bulk</b>     | <b>bullet</b>   | <b>bundle</b>  | <b>bunker</b>   | <b>burden</b>  | <b>burger</b>   | <b>burst</b>    | <b>bus</b>      | <b>business</b> | <b>busy</b>     | <b>butter</b>   | <b>buyer</b>    | <b>buzz</b>     | <b>cabbage</b>  | <b>cabin</b>    | <b>cable</b>   |
| ●○○○○ 17 | <b>cactus</b>   | <b>cage</b>     | <b>cake</b>    | <b>call</b>     | <b>calm</b>    | <b>camera</b>   | <b>camp</b>     | <b>can</b>      | <b>canal</b>    | <b>cancel</b>   | <b>candy</b>    | <b>cannon</b>   | <b>canoe</b>    | <b>canvas</b>   | <b>canyon</b>   | <b>capable</b> |
| ●○○○○ 18 | <b>capital</b>  | <b>captain</b>  | <b>car</b>     | <b>carbon</b>   | <b>card</b>    | <b>cargo</b>    | <b>carpet</b>   | <b>carry</b>    | <b>cart</b>     | <b>case</b>     | <b>cash</b>     | <b>casino</b>   | <b>castle</b>   | <b>casual</b>   | <b>cat</b>      | <b>catalog</b> |
| ●○○○○ 19 | <b>catch</b>    | <b>category</b> | <b>cattle</b>  | <b>caught</b>   | <b>cause</b>   | <b>caution</b>  | <b>cave</b>     | <b>ceiling</b>  | <b>celery</b>   | <b>cement</b>   | <b>census</b>   | <b>century</b>  | <b>cereal</b>   | <b>certain</b>  | <b>chair</b>    | <b>chalk</b>   |
| ●○○○○ 20 | <b>champion</b> | <b>change</b>   | <b>chaos</b>   | <b>chapter</b>  | <b>charge</b>  | <b>chase</b>    | <b>chat</b>     | <b>cheap</b>    | <b>check</b>    | <b>cheese</b>   | <b>chef</b>     | <b>cherry</b>   | <b>chest</b>    | <b>chicken</b>  | <b>chief</b>    | <b>child</b>   |
| ●○○○○ 21 | <b>chimney</b>  | <b>choice</b>   | <b>choose</b>  | <b>chronic</b>  | <b>chuckle</b> | <b>chunk</b>    | <b>churn</b>    | <b>cigar</b>    | <b>cinnamon</b> | <b>circle</b>   | <b>citizen</b>  | <b>city</b>     | <b>civil</b>    | <b>claim</b>    | <b>clap</b>     | <b>clarify</b> |
| ●○○○○ 22 | <b>claw</b>     | <b>clay</b>     | <b>clean</b>   | <b>clerk</b>    | <b>clever</b>  | <b>click</b>    | <b>client</b>   | <b>cliff</b>    | <b>climb</b>    | <b>clinic</b>   | <b>clip</b>     | <b>clock</b>    | <b>clog</b>     | <b>close</b>    | <b>cloth</b>    | <b>cloud</b>   |
| ●○○○○ 23 | <b>clown</b>    | <b>club</b>     | <b>clump</b>   | <b>cluster</b>  | <b>clutch</b>  | <b>coach</b>    | <b>coast</b>    | <b>coconut</b>  | <b>code</b>     | <b>coffee</b>   | <b>coil</b>     | <b>coin</b>     | <b>collect</b>  | <b>color</b>    | <b>column</b>   | <b>combine</b> |
| ●○○○○ 24 | <b>come</b>     | <b>comfort</b>  | <b>comic</b>   | <b>common</b>   | <b>company</b> | <b>concert</b>  | <b>conduct</b>  | <b>confirm</b>  | <b>Congress</b> | <b>connect</b>  | <b>consider</b> | <b>control</b>  | <b>convince</b> | <b>cook</b>     | <b>cool</b>     | <b>copper</b>  |
| ●●○○○ 25 | <b>copy</b>     | <b>coral</b>    | <b>core</b>    | <b>corn</b>     | <b>correct</b> | <b>cost</b>     | <b>cotton</b>   | <b>couch</b>    | <b>country</b>  | <b>couple</b>   | <b>course</b>   | <b>cousin</b>   | <b>cover</b>    | <b>coyote</b>   | <b>crack</b>    | <b>cradle</b>  |
| ●●○○○ 26 | <b>craft</b>    | <b>cram</b>     | <b>crane</b>   | <b>crash</b>    | <b>crater</b>  | <b>crawl</b>    | <b>crazy</b>    | <b>cream</b>    | <b>credit</b>   | <b>creek</b>    | <b>crew</b>     | <b>cricket</b>  | <b>crime</b>    | <b>crisp</b>    | <b>critic</b>   | <b>crop</b>    |
| ●●○○○ 27 | <b>cross</b>    | <b>crouch</b>   | <b>crowd</b>   | <b>crucial</b>  | <b>cruel</b>   | <b>cruise</b>   | <b>crumble</b>  | <b>crunch</b>   | <b>crush</b>    | <b>cry</b>      | <b>crystal</b>  | <b>cube</b>     | <b>culture</b>  | <b>cup</b>      | <b>cupboard</b> | <b>curious</b> |
| ●●○○○ 28 | <b>current</b>  | <b>curtain</b>  | <b>curve</b>   | <b>cushion</b>  | <b>custom</b>  | <b>cute</b>     | <b>cycle</b>    | <b>dad</b>      | <b>damage</b>   | <b>damp</b>     | <b>dance</b>    | <b>danger</b>   | <b>daring</b>   | <b>dash</b>     | <b>daughter</b> | <b>dawn</b>    |
| ●●○○○ 29 | <b>day</b>      | <b>deal</b>     | <b>debate</b>  | <b>debris</b>   | <b>decade</b>  | <b>december</b> | <b>decide</b>   | <b>decline</b>  | <b>decorate</b> | <b>decrease</b> | <b>deer</b>     | <b>defense</b>  | <b>define</b>   | <b>defy</b>     | <b>degree</b>   | <b>delay</b>   |
| ●●○○○ 30 | <b>deliver</b>  | <b>demand</b>   | <b>demise</b>  | <b>denial</b>   | <b>dentist</b> | <b>deny</b>     | <b>depart</b>   | <b>depend</b>   | <b>deposit</b>  | <b>depth</b>    | <b>deputy</b>   | <b>derive</b>   | <b>describe</b> | <b>desert</b>   | <b>design</b>   | <b>desk</b>    |
| ●●○○○ 31 | <b>despair</b>  | <b>destroy</b>  | <b>detail</b>  | <b>detect</b>   | <b>develop</b> | <b>device</b>   | <b>devote</b>   | <b>diagram</b>  | <b>dial</b>     | <b>diamond</b>  | <b>diary</b>    | <b>dice</b>     | <b>diesel</b>   | <b>diet</b>     | <b>differ</b>   | <b>digital</b> |
| ●●○○○ 32 | <b>dignity</b>  | <b>dilemma</b>  | <b>dinner</b>  | <b>dinosaur</b> | <b>direct</b>  | <b>dirt</b>     | <b>disagree</b> | <b>discover</b> | <b>disease</b>  | <b>dish</b>     | <b>dismiss</b>  | <b>disorder</b> | <b>display</b>  | <b>distance</b> | <b>divert</b>   | <b>divide</b>  |



# B



## BIP39 word sheet B of 4 · words 513–1024 of 2,048, in official order

Each die is a coin. Read the top face only: ● **dots**, ○ **blank** (ordinary dice: even ●, odd ○). Every word is equally likely. The first four letters (bold) identify every word uniquely.

**Dice:** sheet **2 orange** | row **5 white** | column **4 black** · within each colour, fewest dots on the left

| row      | ○○○○<br>1       | ○○○●<br>2      | ○○●○<br>3       | ○○●●<br>4       | ○●○○<br>5               | ○●●●<br>6       | ○●●○<br>7      | ○●●●<br>8       | ●○○○<br>9      | ●○○●<br>10      | ●○●○<br>11      | ●○●●<br>12      | ●●○○<br>13      | ●●○●<br>14      | ●●●○<br>15      | ●●●●<br>16      |
|----------|-----------------|----------------|-----------------|-----------------|-------------------------|-----------------|----------------|-----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| ○○○○○ 1  | <b>divorce</b>  | <b>dizzy</b>   | <b>doctor</b>   | <b>document</b> | <b>dog</b>              | <b>doll</b>     | <b>dolphin</b> | <b>domain</b>   | <b>donate</b>  | <b>donkey</b>   | <b>donor</b>    | <b>door</b>     | <b>dose</b>     | <b>double</b>   | <b>dove</b>     | <b>draft</b>    |
| ○○○○○ 2  | <b>dragon</b>   | <b>drama</b>   | <b>drastic</b>  | <b>draw</b>     | <b>dream</b>            | <b>dress</b>    | <b>drift</b>   | <b>drill</b>    | <b>drink</b>   | <b>drip</b>     | <b>drive</b>    | <b>drop</b>     | <b>drum</b>     | <b>dry</b>      | <b>duck</b>     | <b>dumb</b>     |
| ○○○○○ 3  | <b>dune</b>     | <b>during</b>  | <b>dust</b>     | <b>dutch</b>    | <b>duty</b>             | <b>dwarf</b>    | <b>dynamic</b> | <b>eager</b>    | <b>eagle</b>   | <b>early</b>    | <b>earn</b>     | <b>earth</b>    | <b>easily</b>   | <b>east</b>     | <b>easy</b>     | <b>echo</b>     |
| ○○○○○ 4  | <b>ecology</b>  | <b>economy</b> | <b>edge</b>     | <b>edit</b>     | <b>educate</b>          | <b>effort</b>   | <b>egg</b>     | <b>eight</b>    | <b>either</b>  | <b>elbow</b>    | <b>elder</b>    | <b>electric</b> | <b>elegant</b>  | <b>element</b>  | <b>elephant</b> | <b>elevator</b> |
| ○○○○○ 5  | <b>elite</b>    | <b>else</b>    | <b>embark</b>   | <b>embody</b>   | <b>embrace</b>          | <b>emerge</b>   | <b>emotion</b> | <b>employ</b>   | <b>empower</b> | <b>empty</b>    | <b>enable</b>   | <b>enact</b>    | <b>end</b>      | <b>endless</b>  | <b>endorse</b>  | <b>enemy</b>    |
| ○○○○○ 6  | <b>energy</b>   | <b>enforce</b> | <b>engage</b>   | <b>engine</b>   | <b>enhance</b>          | <b>enjoy</b>    | <b>enlist</b>  | <b>enough</b>   | <b>enrich</b>  | <b>enroll</b>   | <b>ensure</b>   | <b>enter</b>    | <b>entire</b>   | <b>entry</b>    | <b>envelope</b> | <b>episode</b>  |
| ○○○○○ 7  | <b>equal</b>    | <b>equip</b>   | <b>era</b>      | <b>erase</b>    | <b>erode</b>            | <b>erosion</b>  | <b>error</b>   | <b>erupt</b>    | <b>escape</b>  | <b>essay</b>    | <b>essence</b>  | <b>estate</b>   | <b>eternal</b>  | <b>ethics</b>   | <b>evidence</b> | <b>evil</b>     |
| ○○○○○ 8  | <b>evoke</b>    | <b>evolve</b>  | <b>exact</b>    | <b>example</b>  | <b>excess</b>           | <b>exchange</b> | <b>excite</b>  | <b>exclude</b>  | <b>excuse</b>  | <b>execute</b>  | <b>exercise</b> | <b>exhaust</b>  | <b>exhibit</b>  | <b>exile</b>    | <b>exist</b>    | <b>exit</b>     |
| ○○○○○ 9  | <b>exotic</b>   | <b>expand</b>  | <b>expect</b>   | <b>expire</b>   | <b>explain</b>          | <b>expose</b>   | <b>express</b> | <b>extend</b>   | <b>extra</b>   | <b>eye</b>      | <b>eyebrow</b>  | <b>fabric</b>   | <b>face</b>     | <b>faculty</b>  | <b>fade</b>     | <b>faint</b>    |
| ○○○○○ 10 | <b>faith</b>    | <b>fall</b>    | <b>false</b>    | <b>fame</b>     | <b>family</b>           | <b>famous</b>   | <b>fan</b>     | <b>fancy</b>    | <b>fantasy</b> | <b>farm</b>     | <b>fashion</b>  | <b>fat</b>      | <b>fatal</b>    | <b>father</b>   | <b>fatigue</b>  | <b>fault</b>    |
| ○○○○○ 11 | <b>favorite</b> | <b>feature</b> | <b>february</b> | <b>federal</b>  | <b>fee</b>              | <b>feed</b>     | <b>feel</b>    | <b>female</b>   | <b>fence</b>   | <b>festival</b> | <b>fetch</b>    | <b>fever</b>    | <b>few</b>      | <b>fiber</b>    | <b>fiction</b>  | <b>field</b>    |
| ○○○○○ 12 | <b>figure</b>   | <b>file</b>    | <b>film</b>     | <b>filter</b>   | <b>final</b>            | <b>find</b>     | <b>fine</b>    | <b>finger</b>   | <b>finish</b>  | <b>fire</b>     | <b>firm</b>     | <b>first</b>    | <b>fiscal</b>   | <b>fish</b>     | <b>fit</b>      | <b>fitness</b>  |
| ○○○○○ 13 | <b>fix</b>      | <b>flag</b>    | <b>flame</b>    | <b>flash</b>    | <b>flat</b>             | <b>flavor</b>   | <b>flee</b>    | <b>flight</b>   | <b>flip</b>    | <b>float</b>    | <b>flock</b>    | <b>floor</b>    | <b>flower</b>   | <b>fluid</b>    | <b>flush</b>    | <b>fly</b>      |
| ○○○○○ 14 | <b>foam</b>     | <b>focus</b>   | <b>fog</b>      | <b>foil</b>     | <b>fold</b>             | <b>follow</b>   | <b>food</b>    | <b>foot</b>     | <b>force</b>   | <b>forest</b>   | <b>forget</b>   | <b>fork</b>     | <b>fortune</b>  | <b>forum</b>    | <b>forward</b>  | <b>fossil</b>   |
| ○○○○○ 15 | <b>foster</b>   | <b>found</b>   | <b>fox</b>      | <b>fragile</b>  | <b>frame</b>            | <b>frequent</b> | <b>fresh</b>   | <b>friend</b>   | <b>fringe</b>  | <b>frog</b>     | <b>front</b>    | <b>frost</b>    | <b>frown</b>    | <b>frozen</b>   | <b>fruit</b>    | <b>fuel</b>     |
| ○○○○○ 16 | <b>fun</b>      | <b>funny</b>   | <b>furnace</b>  | <b>fury</b>     | <b>future</b>           | <b>gadget</b>   | <b>gain</b>    | <b>galaxy</b>   | <b>gallery</b> | <b>game</b>     | <b>gap</b>      | <b>garage</b>   | <b>garbage</b>  | <b>garden</b>   | <b>garlic</b>   | <b>garment</b>  |
| ●○○○○ 17 | <b>gas</b>      | <b>gasp</b>    | <b>gate</b>     | <b>gather</b>   | <b>gauge</b>            | <b>gaze</b>     | <b>general</b> | <b>genius</b>   | <b>genre</b>   | <b>gentle</b>   | <b>genuine</b>  | <b>gesture</b>  | <b>ghost</b>    | <b>giant</b>    | <b>gift</b>     | <b>giggle</b>   |
| ●○○○○ 18 | <b>ginger</b>   | <b>giraffe</b> | <b>girl</b>     | <b>give</b>     | <b>glad</b>             | <b>glance</b>   | <b>glare</b>   | <b>glass</b>    | <b>glide</b>   | <b>glimpse</b>  | <b>globe</b>    | <b>gloom</b>    | <b>glory</b>    | <b>glove</b>    | <b>glow</b>     | <b>glue</b>     |
| ●○○○○ 19 | <b>goat</b>     | <b>goddess</b> | <b>gold</b>     | <b>good</b>     | <b>goose</b>            | <b>gorilla</b>  | <b>gospel</b>  | <b>gossip</b>   | <b>govern</b>  | <b>gown</b>     | <b>grab</b>     | <b>grace</b>    | <b>grain</b>    | <b>grant</b>    | <b>grape</b>    | <b>grass</b>    |
| ●○○○○ 20 | <b>gravity</b>  | <b>great</b>   | <b>green</b>    | <b>grid</b>     | <b>grief</b>            | <b>grit</b>     | <b>grocery</b> | <b>group</b>    | <b>grow</b>    | <b>grunt</b>    | <b>guard</b>    | <b>guess</b>    | <b>guide</b>    | <b>guilt</b>    | <b>guitar</b>   | <b>gun</b>      |
| ●○○○○ 21 | <b>gym</b>      | <b>habit</b>   | <b>hair</b>     | <b>half</b>     | <b>hammer</b>           | <b>hamster</b>  | <b>hand</b>    | <b>happy</b>    | <b>harbor</b>  | <b>hard</b>     | <b>harsh</b>    | <b>harvest</b>  | <b>hat</b>      | <b>have</b>     | <b>hawk</b>     | <b>hazard</b>   |
| ●○○○○ 22 | <b>head</b>     | <b>health</b>  | <b>heart</b>    | <b>heavy</b>    | <b>hedg</b> <b>ehog</b> | <b>height</b>   | <b>hello</b>   | <b>helmet</b>   | <b>help</b>    | <b>hen</b>      | <b>hero</b>     | <b>hidden</b>   | <b>high</b>     | <b>hill</b>     | <b>hint</b>     | <b>hip</b>      |
| ●○○○○ 23 | <b>hire</b>     | <b>history</b> | <b>hobby</b>    | <b>hockey</b>   | <b>hold</b>             | <b>hole</b>     | <b>holiday</b> | <b>hollow</b>   | <b>home</b>    | <b>honey</b>    | <b>hood</b>     | <b>hope</b>     | <b>horn</b>     | <b>horror</b>   | <b>horse</b>    | <b>hospital</b> |
| ●○○○○ 24 | <b>host</b>     | <b>hotel</b>   | <b>hour</b>     | <b>hover</b>    | <b>hub</b>              | <b>huge</b>     | <b>human</b>   | <b>humble</b>   | <b>humor</b>   | <b>hundred</b>  | <b>hungry</b>   | <b>hunt</b>     | <b>hurdle</b>   | <b>hurry</b>    | <b>hurt</b>     | <b>husband</b>  |
| ●●○○○ 25 | <b>hybrid</b>   | <b>ice</b>     | <b>icon</b>     | <b>idea</b>     | <b>identify</b>         | <b>idle</b>     | <b>ignore</b>  | <b>ill</b>      | <b>illegal</b> | <b>illness</b>  | <b>image</b>    | <b>imitate</b>  | <b>immense</b>  | <b>immune</b>   | <b>impact</b>   | <b>impose</b>   |
| ●●○○○ 26 | <b>improve</b>  | <b>impulse</b> | <b>inch</b>     | <b>include</b>  | <b>income</b>           | <b>increase</b> | <b>index</b>   | <b>indicate</b> | <b>indoor</b>  | <b>industry</b> | <b>infant</b>   | <b>inflict</b>  | <b>inform</b>   | <b>inhale</b>   | <b>inherit</b>  | <b>initial</b>  |
| ●●○○○ 27 | <b>inject</b>   | <b>injury</b>  | <b>inmate</b>   | <b>inner</b>    | <b>innocent</b>         | <b>input</b>    | <b>inquiry</b> | <b>insane</b>   | <b>insect</b>  | <b>inside</b>   | <b>inspire</b>  | <b>install</b>  | <b>intact</b>   | <b>interest</b> | <b>into</b>     | <b>invest</b>   |
| ●●○○○ 28 | <b>invite</b>   | <b>involve</b> | <b>iron</b>     | <b>island</b>   | <b>isolate</b>          | <b>issue</b>    | <b>item</b>    | <b>ivory</b>    | <b>jacket</b>  | <b>jaguar</b>   | <b>jar</b>      | <b>jazz</b>     | <b>jealous</b>  | <b>jeans</b>    | <b>jelly</b>    | <b>jewel</b>    |
| ●●○○○ 29 | <b>job</b>      | <b>join</b>    | <b>joke</b>     | <b>journey</b>  | <b>joy</b>              | <b>judge</b>    | <b>juice</b>   | <b>jump</b>     | <b>jungle</b>  | <b>junior</b>   | <b>junk</b>     | <b>just</b>     | <b>kangaroo</b> | <b>keen</b>     | <b>keep</b>     | <b>ketchup</b>  |
| ●●○○○ 30 | <b>key</b>      | <b>kick</b>    | <b>kid</b>      | <b>kidney</b>   | <b>kind</b>             | <b>kingdom</b>  | <b>kiss</b>    | <b>kit</b>      | <b>kitchen</b> | <b>kite</b>     | <b>kitten</b>   | <b>kiwi</b>     | <b>knee</b>     | <b>knife</b>    | <b>knock</b>    | <b>know</b>     |
| ●●○○○ 31 | <b>lab</b>      | <b>label</b>   | <b>labor</b>    | <b>ladder</b>   | <b>lady</b>             | <b>lake</b>     | <b>lamp</b>    | <b>language</b> | <b>laptop</b>  | <b>large</b>    | <b>later</b>    | <b>latin</b>    | <b>laugh</b>    | <b>laundry</b>  | <b>lava</b>     | <b>law</b>      |
| ●●○○○ 32 | <b>lawn</b>     | <b>lawsuit</b> | <b>layer</b>    | <b>lazy</b>     | <b>leader</b>           | <b>leaf</b>     | <b>learn</b>   | <b>leave</b>    | <b>lecture</b> | <b>left</b>     | <b>leg</b>      | <b>legal</b>    | <b>legend</b>   | <b>leisure</b>  | <b>lemon</b>    | <b>lend</b>     |



## BIP39 word sheet C of 4 · words 1025–1536 of 2,048, in official order

Each die is a coin. Read the top face only: ● **dots**, ○ **blank** (ordinary dice: even ●, odd ○). Every word is equally likely. The first four letters (**bold**) identify every word uniquely.

**Dice:** sheet **2 orange** | row **5 white** | column **4 black** · within each colour, fewest dots on the left

| row     | oooo<br>1       | oo●●<br>2       | oo●○<br>3       | oo●●<br>4       | o●○○<br>5       | o●●●<br>6       | o●○○<br>7      | o●●●<br>8       | ●○○○<br>9      | ●○○●<br>10     | ●○○○<br>11      | ●○○●<br>12      | ●●○○<br>13      | ●●○○<br>14      | ●●○○<br>15      | ●●●●<br>16      |
|---------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|----------------|-----------------|----------------|----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| oooo 1  | <b>length</b>   | <b>lens</b>     | <b>leopard</b>  | <b>lesson</b>   | <b>letter</b>   | <b>level</b>    | <b>liar</b>    | <b>liberty</b>  | <b>library</b> | <b>license</b> | <b>life</b>     | <b>lift</b>     | <b>light</b>    | <b>like</b>     | <b>limb</b>     | <b>limit</b>    |
| oooo 2  | <b>link</b>     | <b>lion</b>     | <b>liquid</b>   | <b>list</b>     | <b>little</b>   | <b>live</b>     | <b>lizard</b>  | <b>load</b>     | <b>loan</b>    | <b>lobster</b> | <b>local</b>    | <b>lock</b>     | <b>logic</b>    | <b>lonely</b>   | <b>long</b>     | <b>loop</b>     |
| oooo 3  | <b>lottery</b>  | <b>loud</b>     | <b>lounge</b>   | <b>love</b>     | <b>loyal</b>    | <b>lucky</b>    | <b>luggage</b> | <b>lumber</b>   | <b>lunar</b>   | <b>lunch</b>   | <b>luxury</b>   | <b>lyrics</b>   | <b>machine</b>  | <b>mad</b>      | <b>magic</b>    | <b>magnet</b>   |
| oooo 4  | <b>maid</b>     | <b>mail</b>     | <b>main</b>     | <b>major</b>    | <b>make</b>     | <b>mammal</b>   | <b>man</b>     | <b>manage</b>   | <b>mandate</b> | <b>mango</b>   | <b>mansion</b>  | <b>manual</b>   | <b>maple</b>    | <b>marble</b>   | <b>march</b>    | <b>margin</b>   |
| oooo 5  | <b>marine</b>   | <b>market</b>   | <b>marriage</b> | <b>mask</b>     | <b>mass</b>     | <b>master</b>   | <b>match</b>   | <b>material</b> | <b>math</b>    | <b>matrix</b>  | <b>matter</b>   | <b>maximum</b>  | <b>maze</b>     | <b>meadow</b>   | <b>mean</b>     | <b>measure</b>  |
| oooo 6  | <b>meat</b>     | <b>mechanic</b> | <b>medal</b>    | <b>media</b>    | <b>melody</b>   | <b>melt</b>     | <b>member</b>  | <b>memory</b>   | <b>mention</b> | <b>menu</b>    | <b>mercy</b>    | <b>merge</b>    | <b>merit</b>    | <b>merry</b>    | <b>mesh</b>     | <b>message</b>  |
| oooo 7  | <b>metal</b>    | <b>method</b>   | <b>middle</b>   | <b>midnight</b> | <b>milk</b>     | <b>million</b>  | <b>mimic</b>   | <b>mind</b>     | <b>minimum</b> | <b>minor</b>   | <b>minute</b>   | <b>miracle</b>  | <b>mirror</b>   | <b>misery</b>   | <b>miss</b>     | <b>mistake</b>  |
| oooo 8  | <b>mix</b>      | <b>mixed</b>    | <b>mixture</b>  | <b>mobile</b>   | <b>model</b>    | <b>modify</b>   | <b>mom</b>     | <b>moment</b>   | <b>monitor</b> | <b>monkey</b>  | <b>monster</b>  | <b>month</b>    | <b>moon</b>     | <b>moral</b>    | <b>more</b>     | <b>morning</b>  |
| oooo 9  | <b>mosquito</b> | <b>mother</b>   | <b>motion</b>   | <b>motor</b>    | <b>mountain</b> | <b>mouse</b>    | <b>move</b>    | <b>movie</b>    | <b>much</b>    | <b>muffin</b>  | <b>mule</b>     | <b>multiply</b> | <b>muscle</b>   | <b>museum</b>   | <b>mushroom</b> | <b>music</b>    |
| oooo 10 | <b>must</b>     | <b>mutual</b>   | <b>myself</b>   | <b>mystery</b>  | <b>myth</b>     | <b>naive</b>    | <b>name</b>    | <b>napkin</b>   | <b>narrow</b>  | <b>nasty</b>   | <b>nation</b>   | <b>nature</b>   | <b>near</b>     | <b>neck</b>     | <b>need</b>     | <b>negative</b> |
| oooo 11 | <b>neglect</b>  | <b>neither</b>  | <b>nephew</b>   | <b>nerve</b>    | <b>nest</b>     | <b>net</b>      | <b>network</b> | <b>neutral</b>  | <b>never</b>   | <b>news</b>    | <b>next</b>     | <b>nice</b>     | <b>night</b>    | <b>noble</b>    | <b>noise</b>    | <b>nominee</b>  |
| oooo 12 | <b>noodle</b>   | <b>normal</b>   | <b>north</b>    | <b>nose</b>     | <b>notable</b>  | <b>note</b>     | <b>nothing</b> | <b>notice</b>   | <b>novel</b>   | <b>now</b>     | <b>nuclear</b>  | <b>number</b>   | <b>nurse</b>    | <b>nut</b>      | <b>oak</b>      | <b>obey</b>     |
| oooo 13 | <b>object</b>   | <b>oblige</b>   | <b>obscure</b>  | <b>observe</b>  | <b>obtain</b>   | <b>obvious</b>  | <b>occur</b>   | <b>ocean</b>    | <b>october</b> | <b>odor</b>    | <b>off</b>      | <b>offer</b>    | <b>office</b>   | <b>often</b>    | <b>oil</b>      | <b>okay</b>     |
| oooo 14 | <b>old</b>      | <b>olive</b>    | <b>olympic</b>  | <b>omit</b>     | <b>once</b>     | <b>one</b>      | <b>onion</b>   | <b>online</b>   | <b>only</b>    | <b>open</b>    | <b>opera</b>    | <b>opinion</b>  | <b>oppose</b>   | <b>option</b>   | <b>orange</b>   | <b>orbit</b>    |
| oooo 15 | <b>orchard</b>  | <b>order</b>    | <b>ordinary</b> | <b>organ</b>    | <b>orient</b>   | <b>original</b> | <b>orphan</b>  | <b>ostrich</b>  | <b>other</b>   | <b>outdoor</b> | <b>outer</b>    | <b>output</b>   | <b>outside</b>  | <b>oval</b>     | <b>oven</b>     | <b>over</b>     |
| oooo 16 | <b>own</b>      | <b>owner</b>    | <b>oxygen</b>   | <b>oyster</b>   | <b>ozone</b>    | <b>pact</b>     | <b>paddle</b>  | <b>page</b>     | <b>pair</b>    | <b>palace</b>  | <b>palm</b>     | <b>panda</b>    | <b>panel</b>    | <b>panic</b>    | <b>panther</b>  | <b>paper</b>    |
| oooo 17 | <b>parade</b>   | <b>parent</b>   | <b>park</b>     | <b>parrot</b>   | <b>party</b>    | <b>pass</b>     | <b>patch</b>   | <b>path</b>     | <b>patient</b> | <b>patrol</b>  | <b>pattern</b>  | <b>pause</b>    | <b>pave</b>     | <b>payment</b>  | <b>peace</b>    | <b>peanut</b>   |
| oooo 18 | <b>pear</b>     | <b>peasant</b>  | <b>pelican</b>  | <b>pen</b>      | <b>penalty</b>  | <b>pencil</b>   | <b>people</b>  | <b>pepper</b>   | <b>perfect</b> | <b>permit</b>  | <b>person</b>   | <b>pet</b>      | <b>phone</b>    | <b>photo</b>    | <b>phrase</b>   | <b>physical</b> |
| oooo 19 | <b>piano</b>    | <b>picnic</b>   | <b>picture</b>  | <b>piece</b>    | <b>pig</b>      | <b>pigeon</b>   | <b>pill</b>    | <b>pilot</b>    | <b>pink</b>    | <b>pioneer</b> | <b>pipe</b>     | <b>pistol</b>   | <b>pitch</b>    | <b>pizza</b>    | <b>place</b>    | <b>planet</b>   |
| oooo 20 | <b>plastic</b>  | <b>plate</b>    | <b>play</b>     | <b>please</b>   | <b>pledge</b>   | <b>pluck</b>    | <b>plug</b>    | <b>plunge</b>   | <b>poem</b>    | <b>poet</b>    | <b>point</b>    | <b>polar</b>    | <b>pole</b>     | <b>police</b>   | <b>pond</b>     | <b>pony</b>     |
| oooo 21 | <b>pool</b>     | <b>popular</b>  | <b>portion</b>  | <b>position</b> | <b>possible</b> | <b>post</b>     | <b>potato</b>  | <b>pottery</b>  | <b>poverty</b> | <b>powder</b>  | <b>power</b>    | <b>practice</b> | <b>praise</b>   | <b>predict</b>  | <b>prefer</b>   | <b>prepare</b>  |
| oooo 22 | <b>present</b>  | <b>pretty</b>   | <b>prevent</b>  | <b>price</b>    | <b>pride</b>    | <b>primary</b>  | <b>print</b>   | <b>priority</b> | <b>prison</b>  | <b>private</b> | <b>prize</b>    | <b>problem</b>  | <b>process</b>  | <b>produce</b>  | <b>profit</b>   | <b>program</b>  |
| oooo 23 | <b>project</b>  | <b>promote</b>  | <b>proof</b>    | <b>property</b> | <b>prosper</b>  | <b>protect</b>  | <b>proud</b>   | <b>provide</b>  | <b>public</b>  | <b>pudding</b> | <b>pull</b>     | <b>pulp</b>     | <b>pulse</b>    | <b>pumpkin</b>  | <b>punch</b>    | <b>pupil</b>    |
| oooo 24 | <b>puppy</b>    | <b>purchase</b> | <b>purity</b>   | <b>purpose</b>  | <b>purse</b>    | <b>push</b>     | <b>put</b>     | <b>puzzle</b>   | <b>pyramid</b> | <b>quality</b> | <b>quantum</b>  | <b>quarter</b>  | <b>question</b> | <b>quick</b>    | <b>quit</b>     | <b>quiz</b>     |
| oooo 25 | <b>quote</b>    | <b>rabbit</b>   | <b>raccoon</b>  | <b>race</b>     | <b>rack</b>     | <b>radar</b>    | <b>radio</b>   | <b>rail</b>     | <b>rain</b>    | <b>raise</b>   | <b>rally</b>    | <b>ramp</b>     | <b>ranch</b>    | <b>random</b>   | <b>range</b>    | <b>rapid</b>    |
| oooo 26 | <b>rare</b>     | <b>rate</b>     | <b>rather</b>   | <b>raven</b>    | <b>raw</b>      | <b>razor</b>    | <b>ready</b>   | <b>real</b>     | <b>reason</b>  | <b>rebel</b>   | <b>rebuild</b>  | <b>recall</b>   | <b>receive</b>  | <b>recipe</b>   | <b>record</b>   | <b>recycle</b>  |
| oooo 27 | <b>reduce</b>   | <b>reflect</b>  | <b>reform</b>   | <b>refuse</b>   | <b>region</b>   | <b>regret</b>   | <b>regular</b> | <b>reject</b>   | <b>relax</b>   | <b>release</b> | <b>relief</b>   | <b>rely</b>     | <b>remain</b>   | <b>remember</b> | <b>remind</b>   | <b>remove</b>   |
| oooo 28 | <b>render</b>   | <b>renew</b>    | <b>rent</b>     | <b>reopen</b>   | <b>repair</b>   | <b>repeat</b>   | <b>replace</b> | <b>report</b>   | <b>require</b> | <b>rescue</b>  | <b>resemble</b> | <b>resist</b>   | <b>resource</b> | <b>response</b> | <b>result</b>   | <b>retire</b>   |
| oooo 29 | <b>retreat</b>  | <b>return</b>   | <b>reunion</b>  | <b>reveal</b>   | <b>review</b>   | <b>reward</b>   | <b>rhythm</b>  | <b>rib</b>      | <b>ribbon</b>  | <b>rice</b>    | <b>rich</b>     | <b>ride</b>     | <b>ridge</b>    | <b>rifle</b>    | <b>right</b>    | <b>rigid</b>    |
| oooo 30 | <b>ring</b>     | <b>riot</b>     | <b>ripple</b>   | <b>risk</b>     | <b>ritual</b>   | <b>rival</b>    | <b>river</b>   | <b>road</b>     | <b>roast</b>   | <b>robot</b>   | <b>robust</b>   | <b>rocket</b>   | <b>romance</b>  | <b>roof</b>     | <b>rookie</b>   | <b>room</b>     |
| oooo 31 | <b>rose</b>     | <b>rotate</b>   | <b>rough</b>    | <b>round</b>    | <b>route</b>    | <b>royal</b>    | <b>rubber</b>  | <b>rude</b>     | <b>rug</b>     | <b>rule</b>    | <b>run</b>      | <b>runway</b>   | <b>rural</b>    | <b>sad</b>      | <b>saddle</b>   | <b>sadness</b>  |
| oooo 32 | <b>safe</b>     | <b>sail</b>     | <b>salad</b>    | <b>salmon</b>   | <b>salon</b>    | <b>salt</b>     | <b>salute</b>  | <b>same</b>     | <b>sample</b>  | <b>sand</b>    | <b>satisfy</b>  | <b>satoshi</b>  | <b>sauce</b>    | <b>sausage</b>  | <b>save</b>     | <b>say</b>      |



## BIP39 word sheet D of 4 · words 1537–2048 of 2,048, in official order

Each die is a coin. Read the top face only: ● **dots**, ○ **blank** (ordinary dice: even ●, odd ○). Every word is equally likely. The first four letters (**bold**) identify every word uniquely.

**Dice:** sheet **2 orange** | row **5 white** | column **4 black** · within each colour, fewest dots on the left

| row     | oooo<br>1                                             | oo●●<br>2                                              | oo●○<br>3                                            | oo●●<br>4                                     | o●○○<br>5                                    | o●●●<br>6                                            | o●○○<br>7                                              | o●●●<br>8                                    | ●○○○<br>9                                    | ●○○●<br>10                                                      | ●○●○<br>11                                   | ●○●●<br>12                                    | ●●○○<br>13                                   | ●●○●<br>14                                            | ●●○○<br>15                                             | ●●●●<br>16                                    |
|---------|-------------------------------------------------------|--------------------------------------------------------|------------------------------------------------------|-----------------------------------------------|----------------------------------------------|------------------------------------------------------|--------------------------------------------------------|----------------------------------------------|----------------------------------------------|-----------------------------------------------------------------|----------------------------------------------|-----------------------------------------------|----------------------------------------------|-------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------|
| oooo 1  | <b>s</b> cale                                         | <b>s</b> can                                           | <b>s</b> care                                        | <b>s</b> catter                               | <b>s</b> cene                                | <b>s</b> cheme                                       | <b>s</b> chool                                         | <b>s</b> cience                              | <b>s</b> cissors                             | <b>s</b> corpion                                                | <b>s</b> cout                                | <b>s</b> crap                                 | <b>s</b> creen                               | <b>s</b> cript                                        | <b>s</b> crub                                          | <b>s</b> ea                                   |
| oooo 2  | <b>s</b> earch                                        | <b>s</b> ea <b>son</b>                                 | <b>s</b> ea <b>t</b>                                 | <b>s</b> ea <b>cond</b>                       | <b>s</b> ecret                               | <b>s</b> ection                                      | <b>s</b> ecur <b>ity</b>                               | <b>s</b> eed                                 | <b>s</b> ee <b>k</b>                         | <b>s</b> eg <b>ment</b>                                         | <b>s</b> ele <b>ct</b>                       | <b>s</b> ell                                  | <b>s</b> eminar                              | <b>s</b> enior                                        | <b>s</b> ense                                          | <b>s</b> ent <b>ence</b>                      |
| oooo 3  | <b>s</b> erie <b>s</b>                                | <b>s</b> er <b>vice</b>                                | <b>s</b> ess <b>ion</b>                              | <b>s</b> ett <b>le</b>                        | <b>s</b> et <b>up</b>                        | <b>s</b> ev <b>en</b>                                | <b>s</b> hadow                                         | <b>s</b> h <b>aft</b>                        | <b>s</b> hallow                              | <b>s</b> h <b>are</b>                                           | <b>s</b> h <b>ed</b>                         | <b>s</b> h <b>ell</b>                         | <b>s</b> heriff                              | <b>s</b> hield                                        | <b>s</b> h <b>ift</b>                                  | <b>s</b> h <b>ine</b>                         |
| oooo 4  | <b>s</b> h <b>ip</b>                                  | <b>s</b> h <b>iver</b>                                 | <b>s</b> h <b>ock</b>                                | <b>s</b> h <b>oe</b>                          | <b>s</b> h <b>oot</b>                        | <b>s</b> h <b>op</b>                                 | <b>s</b> h <b>ort</b>                                  | <b>s</b> h <b>oulder</b>                     | <b>s</b> h <b>ove</b>                        | <b>s</b> h <b>rimp</b>                                          | <b>s</b> h <b>rug</b>                        | <b>s</b> h <b>uffle</b>                       | <b>s</b> h <b>y</b>                          | <b>s</b> ib <b>ling</b>                               | <b>s</b> ick                                           | <b>s</b> ide                                  |
| oooo 5  | <b>s</b> iege                                         | <b>s</b> igh <b>t</b>                                  | <b>s</b> igh <b>l</b> y                              | <b>s</b> il <b>ent</b>                        | <b>s</b> ilk                                 | <b>s</b> illy                                        | <b>s</b> il <b>ver</b>                                 | <b>s</b> im <b>ilar</b>                      | <b>s</b> im <b>ple</b>                       | <b>s</b> in <b>ce</b>                                           | <b>s</b> ing                                 | <b>s</b> ir <b>en</b>                         | <b>s</b> is <b>ter</b>                       | <b>s</b> itua <b>te</b>                               | <b>s</b> ix                                            | <b>s</b> iz <b>e</b>                          |
| oooo 6  | <b>s</b> ka <b>t</b> e                                | <b>s</b> ke <b>t</b> ch                                | <b>s</b> ki                                          | <b>s</b> kil <b>l</b>                         | <b>s</b> kin                                 | <b>s</b> kirt                                        | <b>s</b> ku <b>l</b>                                   | <b>s</b> la <b>b</b>                         | <b>s</b> la <b>m</b>                         | <b>s</b> lee <b>p</b>                                           | <b>s</b> le <b>nd</b> er                     | <b>s</b> li <b>ce</b>                         | <b>s</b> li <b>de</b>                        | <b>s</b> li <b>gh</b> t                               | <b>s</b> li <b>m</b>                                   | <b>s</b> lo <b>ga</b> n                       |
| oooo 7  | <b>s</b> l <b>o</b> t                                 | <b>s</b> l <b>ow</b>                                   | <b>s</b> lu <b>sh</b>                                | <b>s</b> ma <b>ll</b>                         | <b>s</b> ma <b>rt</b>                        | <b>s</b> mi <b>le</b>                                | <b>s</b> mo <b>k</b> e                                 | <b>s</b> mo <b>o</b> th                      | <b>s</b> na <b>ck</b>                        | <b>s</b> na <b>k</b> e                                          | <b>s</b> na <b>p</b>                         | <b>s</b> ni <b>ff</b>                         | <b>s</b> no <b>w</b>                         | <b>s</b> oa <b>p</b>                                  | <b>s</b> occe <b>r</b>                                 | <b>s</b> ocia <b>l</b>                        |
| oooo 8  | <b>s</b> oc <b>k</b>                                  | <b>s</b> o <b>d</b> a                                  | <b>s</b> o <b>ft</b>                                 | <b>s</b> ola <b>r</b>                         | <b>s</b> oldier                              | <b>s</b> oli <b>d</b>                                | <b>s</b> olu <b>ti</b> on                              | <b>s</b> olve                                | <b>s</b> omeone                              | <b>s</b> o <b>ng</b>                                            | <b>s</b> o <b>on</b>                         | <b>s</b> or <b>ry</b>                         | <b>s</b> or <b>t</b>                         | <b>s</b> ou <b>l</b>                                  | <b>s</b> ou <b>nd</b>                                  | <b>s</b> ou <b>p</b>                          |
| oooo 9  | <b>s</b> ou <b>r</b> ce                               | <b>s</b> ou <b>th</b>                                  | <b>s</b> pa <b>ce</b>                                | <b>s</b> pa <b>r</b> e                        | <b>s</b> pa <b>t</b> ial                     | <b>s</b> pa <b>w</b> n                               | <b>s</b> pea <b>k</b>                                  | <b>s</b> pe <b>ci</b> al                     | <b>s</b> pee <b>d</b>                        | <b>s</b> pe <b>ll</b>                                           | <b>s</b> pe <b>nd</b>                        | <b>s</b> pe <b>re</b>                         | <b>s</b> pi <b>ce</b>                        | <b>s</b> pi <b>d</b> er                               | <b>s</b> pi <b>k</b> e                                 | <b>s</b> pi <b>n</b>                          |
| oooo 10 | <b>s</b> pi <b>r</b> it                               | <b>s</b> pli <b>t</b>                                  | <b>s</b> p <i>oi</i> l                               | <b>s</b> pon <b>s</b> or                      | <b>s</b> po <b>o</b> n                       | <b>s</b> po <b>r</b> t                               | <b>s</b> po <b>t</b>                                   | <b>s</b> pra <b>y</b>                        | <b>s</b> prea <b>d</b>                       | <b>s</b> pr <b>ing</b>                                          | <b>s</b> py                                  | <b>s</b> qua <b>r</b> e                       | <b>s</b> que <b>ez</b> e                     | <b>s</b> qui <b>r</b> rel                             | <b>s</b> t <b>ab</b> le                                | <b>s</b> t <b>adi</b> um                      |
| oooo 11 | <b>s</b> ta <b>ff</b>                                 | <b>s</b> t <b>a</b> ge                                 | <b>s</b> t <b>ai</b> rs                              | <b>s</b> t <b>a</b> mp                        | <b>s</b> t <b>a</b> nd                       | <b>s</b> t <b>a</b> rt                               | <b>s</b> t <b>a</b> te                                 | <b>s</b> t <b>a</b> y                        | <b>s</b> t <b>e</b> ak                       | <b>s</b> t <b>e</b> el                                          | <b>s</b> t <b>e</b> m                        | <b>s</b> t <b>e</b> p                         | <b>s</b> t <b>e</b> re <b>o</b>              | <b>s</b> t <b>i</b> ck                                | <b>s</b> t <b>i</b> ll                                 | <b>s</b> t <b>i</b> ng                        |
| oooo 12 | <b>s</b> toc <b>k</b>                                 | <b>s</b> t <b>o</b> mach                               | <b>s</b> t <b>o</b> ne                               | <b>s</b> t <b>oo</b> l                        | <b>s</b> t <b>o</b> ry                       | <b>s</b> t <b>o</b> ve                               | <b>s</b> t <b>r</b> ategy                              | <b>s</b> t <b>r</b> ee <b>t</b>              | <b>s</b> t <b>r</b> i <b>k</b> e             | <b>s</b> t <b>r</b> o <b>ng</b>                                 | <b>s</b> t <b>r</b> ugg <b>le</b>            | <b>s</b> t <b>u</b> dent                      | <b>s</b> t <b>u</b> ff                       | <b>s</b> t <b>u</b> mble                              | <b>s</b> t <b>y</b> le                                 | <b>s</b> ub <b>j</b> ect                      |
| oooo 13 | <b>s</b> ubmit                                        | <b>s</b> ubway                                         | <b>s</b> ucc <b>ess</b>                              | <b>s</b> uch                                  | <b>s</b> udden                               | <b>s</b> uff <b>er</b>                               | <b>s</b> uga <b>r</b>                                  | <b>s</b> ugg <b>est</b>                      | <b>s</b> ui <b>t</b>                         | <b>s</b> um <b>mer</b>                                          | <b>s</b> un                                  | <b>s</b> un <b>ny</b>                         | <b>s</b> un <b>set</b>                       | <b>s</b> u <b>per</b>                                 | <b>s</b> u <b>pp</b> ly                                | <b>s</b> u <b>pre</b> me                      |
| oooo 14 | <b>s</b> u <b>r</b> e                                 | <b>s</b> ur <b>face</b>                                | <b>s</b> ur <b>ge</b>                                | <b>s</b> ur <b>pr</b> ise                     | <b>s</b> urround                             | <b>s</b> ur <b>vey</b>                               | <b>s</b> us <b>pect</b>                                | <b>s</b> ust <b>ain</b>                      | <b>s</b> wa <b>ll</b> ow                     | <b>s</b> wa <b>m</b> p                                          | <b>s</b> wa <b>p</b>                         | <b>s</b> wa <b>r</b> m                        | <b>s</b> wea <b>r</b>                        | <b>s</b> wee <b>t</b>                                 | <b>s</b> wi <b>ft</b>                                  | <b>s</b> wi <b>m</b>                          |
| oooo 15 | <b>s</b> wi <b>ng</b>                                 | <b>s</b> wi <b>t</b> ch                                | <b>s</b> wo <b>rd</b>                                | <b>s</b> ymb <b>ol</b>                        | <b>s</b> ymp <b>to</b> m                     | <b>s</b> yr <b>u</b> p                               | <b>s</b> yst <b>em</b>                                 | <b>s</b> ta <b>bl</b> e                      | <b>s</b> tack <b>le</b>                      | <b>s</b> t <b>a</b> g                                           | <b>s</b> t <b>ai</b> l                       | <b>s</b> t <b>al</b> ent                      | <b>s</b> t <b>a</b> lk                       | <b>s</b> t <b>a</b> nk                                | <b>s</b> t <b>a</b> p <b>e</b>                         | <b>s</b> t <b>a</b> r <b>g</b> et             |
| oooo 16 | <b>s</b> ta <b>s</b> k                                | <b>s</b> t <b>a</b> st <b>e</b>                        | <b>s</b> t <b>a</b> tt <b>oo</b>                     | <b>s</b> t <b>a</b> xi                        | <b>s</b> t <b>e</b> ach                      | <b>s</b> t <b>e</b> am                               | <b>s</b> t <b>e</b> ll                                 | <b>s</b> t <b>e</b> n                        | <b>s</b> t <b>e</b> nant                     | <b>s</b> t <b>e</b> n <b>n</b> is                               | <b>s</b> t <b>e</b> nt                       | <b>s</b> t <b>e</b> r <b>m</b>                | <b>s</b> t <b>e</b> st                       | <b>s</b> t <b>e</b> xt                                | <b>s</b> t <b>a</b> n <b>k</b>                         | <b>s</b> t <b>a</b> t                         |
| oooo 17 | <b>s</b> t <b>e</b> m <b>e</b>                        | <b>s</b> t <b>e</b> n                                  | <b>s</b> t <b>e</b> o <b>r</b> y                     | <b>s</b> t <b>e</b> r <b>e</b>                | <b>s</b> t <b>e</b> y                        | <b>s</b> t <b>i</b> ng                               | <b>s</b> t <b>i</b> s                                  | <b>s</b> t <b>h</b> ought                    | <b>s</b> t <b>h</b> ree                      | <b>s</b> t <b>h</b> rive                                        | <b>s</b> t <b>h</b> row                      | <b>s</b> t <b>h</b> umb                       | <b>s</b> t <b>h</b> under                    | <b>s</b> t <b>i</b> cket                              | <b>s</b> t <b>i</b> d <b>e</b>                         | <b>s</b> t <b>i</b> g <b>e</b> r              |
| oooo 18 | <b>s</b> t <b>i</b> l <b>t</b>                        | <b>s</b> t <b>i</b> mber                               | <b>s</b> t <b>i</b> m <b>e</b>                       | <b>s</b> t <b>i</b> n <b>y</b>                | <b>s</b> t <b>i</b> p                        | <b>s</b> t <b>i</b> red                              | <b>s</b> t <b>i</b> ssue                               | <b>s</b> t <b>i</b> tl <b>e</b>              | <b>s</b> t <b>oa</b> st                      | <b>s</b> t <b>o</b> bacco                                       | <b>s</b> t <b>o</b> da <b>y</b>              | <b>s</b> t <b>o</b> ddler                     | <b>s</b> t <b>o</b> e                        | <b>s</b> t <b>o</b> g <b>e</b> th <b>e</b> r          | <b>s</b> t <b>o</b> il <b>e</b> t                      | <b>s</b> t <b>o</b> k <b>e</b> n              |
| oooo 19 | <b>s</b> t <b>o</b> mato                              | <b>s</b> t <b>o</b> m <b>o</b> rw <b>e</b> d           | <b>s</b> t <b>o</b> n <b>e</b>                       | <b>s</b> t <b>o</b> ng <b>ue</b>              | <b>s</b> t <b>o</b> n <b>i</b> ght           | <b>s</b> t <b>o</b> ol                               | <b>s</b> t <b>o</b> oth                                | <b>s</b> t <b>o</b> p                        | <b>s</b> t <b>o</b> p <b>i</b> c             | <b>s</b> t <b>o</b> pp <b>le</b>                                | <b>s</b> t <b>o</b> r <b>ch</b>              | <b>s</b> t <b>o</b> r <b>n</b> ado            | <b>s</b> t <b>o</b> r <b>t</b> oise          | <b>s</b> t <b>o</b> ss                                | <b>s</b> t <b>o</b> t <b>a</b> l                       | <b>s</b> t <b>o</b> ur <b>i</b> st            |
| oooo 20 | <b>s</b> t <b>o</b> wa <b>r</b> d                     | <b>s</b> t <b>o</b> w <b>e</b> r                       | <b>s</b> t <b>o</b> w <b>n</b>                       | <b>s</b> t <b>o</b> y                         | <b>s</b> t <b>r</b> ack                      | <b>s</b> t <b>r</b> ade                              | <b>s</b> t <b>r</b> aff <b>i</b> c                     | <b>s</b> t <b>r</b> ag <b>i</b> c            | <b>s</b> t <b>r</b> ain                      | <b>s</b> t <b>r</b> ans <b>fer</b>                              | <b>s</b> t <b>r</b> ap                       | <b>s</b> t <b>r</b> ash                       | <b>s</b> t <b>r</b> av <b>e</b> l            | <b>s</b> t <b>r</b> ay                                | <b>s</b> t <b>r</b> ea <b>t</b>                        | <b>s</b> t <b>r</b> ee                        |
| oooo 21 | <b>s</b> t <b>r</b> end                               | <b>s</b> t <b>r</b> ial                                | <b>s</b> t <b>r</b> i <b>b</b> e                     | <b>s</b> t <b>r</b> i <b>k</b> e              | <b>s</b> t <b>r</b> i <b>g</b> ger           | <b>s</b> t <b>r</b> i <b>m</b>                       | <b>s</b> t <b>r</b> i <b>p</b>                         | <b>s</b> t <b>r</b> o <b>ph</b> y            | <b>s</b> t <b>r</b> o <b>u</b> ble           | <b>s</b> t <b>r</b> uck                                         | <b>s</b> t <b>r</b> ue                       | <b>s</b> t <b>r</b> u <b>l</b> y              | <b>s</b> t <b>r</b> ump <b>e</b> t           | <b>s</b> t <b>r</b> ust                               | <b>s</b> t <b>r</b> uth                                | <b>s</b> t <b>r</b> y                         |
| oooo 22 | <b>s</b> t <b>u</b> b <b>e</b>                        | <b>s</b> t <b>u</b> i <b>ti</b> on                     | <b>s</b> t <b>u</b> m <b>bl</b> e                    | <b>s</b> t <b>u</b> na                        | <b>s</b> t <b>u</b> n <b>n</b> el            | <b>s</b> t <b>u</b> r <b>k</b> ey                    | <b>s</b> t <b>u</b> r <b>n</b>                         | <b>s</b> t <b>u</b> r <b>t</b> le            | <b>s</b> t <b>w</b> elve                     | <b>s</b> t <b>w</b> enty                                        | <b>s</b> t <b>w</b> ic <b>e</b>              | <b>s</b> t <b>w</b> i <b>n</b>                | <b>s</b> t <b>w</b> i <b>s</b> t             | <b>s</b> t <b>w</b> o                                 | <b>s</b> t <b>y</b> p <b>e</b>                         | <b>s</b> t <b>y</b> p <b>i</b> c <b>a</b> l   |
| oooo 23 | <b>s</b> t <b>u</b> g <b>l</b> y                      | <b>s</b> t <b>u</b> m <b>br</b> ella                   | <b>s</b> t <b>u</b> n <b>a</b> bl <b>e</b>           | <b>s</b> t <b>u</b> n <b>a</b> wa <b>r</b> e  | <b>s</b> t <b>u</b> n <b>c</b> le            | <b>s</b> t <b>u</b> n <b>c</b> ov <b>e</b> r         | <b>s</b> t <b>u</b> n <b>d</b> e <b>r</b>              | <b>s</b> t <b>u</b> n <b>d</b> o             | <b>s</b> t <b>u</b> n <b>f</b> air           | <b>s</b> t <b>u</b> n <b>f</b> old                              | <b>s</b> t <b>u</b> n <b>h</b> app <b>y</b>  | <b>s</b> t <b>u</b> n <b>f</b> orm            | <b>s</b> t <b>u</b> n <b>i</b> que           | <b>s</b> t <b>u</b> n <b>i</b> t                      | <b>s</b> t <b>u</b> n <b>i</b> v <b>e</b> r <b>s</b> e | <b>s</b> t <b>u</b> n <b>kn</b> own           |
| oooo 24 | <b>s</b> t <b>u</b> n <b>l</b> ock                    | <b>s</b> t <b>u</b> n <b>t</b> i                       | <b>s</b> t <b>u</b> n <b>u</b> s <b>u</b> al         | <b>s</b> t <b>u</b> n <b>v</b> eil            | <b>s</b> t <b>u</b> p <b>d</b> ate           | <b>s</b> t <b>u</b> p <b>gr</b> ade                  | <b>s</b> t <b>u</b> p <b>h</b> old                     | <b>s</b> t <b>u</b> p <b>on</b>              | <b>s</b> t <b>u</b> p <b>p</b> er            | <b>s</b> t <b>u</b> p <b>s</b> e <b>t</b>                       | <b>s</b> t <b>u</b> r <b>ba</b> n            | <b>s</b> t <b>u</b> r <b>g</b> e              | <b>s</b> t <b>u</b> s <b>a</b> g <b>e</b>    | <b>s</b> t <b>u</b> s <b>e</b>                        | <b>s</b> t <b>u</b> s <b>e</b> d                       | <b>s</b> t <b>u</b> s <b>e</b> fu <b>l</b>    |
| oooo 25 | <b>s</b> t <b>u</b> s <b>e</b> l <b>ess</b>           | <b>s</b> t <b>u</b> s <b>u</b> al                      | <b>s</b> t <b>u</b> t <b>i</b> l <b>i</b> ty         | <b>s</b> t <b>u</b> v <b>a</b> nt             | <b>s</b> t <b>u</b> v <b>ac</b> uum          | <b>s</b> t <b>u</b> v <b>ag</b> ue                   | <b>s</b> t <b>u</b> v <b>a</b> l <b>i</b> d            | <b>s</b> t <b>u</b> v <b>al</b> l <b>e</b> y | <b>s</b> t <b>u</b> v <b>al</b> ve           | <b>s</b> t <b>u</b> v <b>a</b> n                                | <b>s</b> t <b>u</b> v <b>a</b> n <b>i</b> sh | <b>s</b> t <b>u</b> v <b>a</b> p <b>or</b>    | <b>s</b> t <b>u</b> v <b>ari</b> ous         | <b>s</b> t <b>u</b> v <b>a</b> st                     | <b>s</b> t <b>u</b> v <b>a</b> ult                     | <b>s</b> t <b>u</b> v <b>e</b> hic <b>l</b> e |
| oooo 26 | <b>s</b> t <b>u</b> v <b>e</b> l <b>e</b> t           | <b>s</b> t <b>u</b> v <b>e</b> n <b>d</b> or           | <b>s</b> t <b>u</b> v <b>e</b> n <b>u</b> r <b>e</b> | <b>s</b> t <b>u</b> v <b>e</b> n <b>ue</b>    | <b>s</b> t <b>u</b> v <b>e</b> r <b>b</b>    | <b>s</b> t <b>u</b> v <b>e</b> r <b>i</b> f <b>y</b> | <b>s</b> t <b>u</b> v <b>e</b> r <b>s</b> i <b>o</b> n | <b>s</b> t <b>u</b> v <b>e</b> r <b>y</b>    | <b>s</b> t <b>u</b> v <b>e</b> s <b>s</b> el | <b>s</b> t <b>u</b> v <b>e</b> r <b>e</b> t <b>a</b> r <b>n</b> | <b>s</b> t <b>u</b> v <b>i</b> abl <b>e</b>  | <b>s</b> t <b>u</b> v <b>i</b> br <b>a</b> nt | <b>s</b> t <b>u</b> v <b>i</b> cious         | <b>s</b> t <b>u</b> v <b>i</b> ct <b>o</b> r <b>y</b> | <b>s</b> t <b>u</b> v <b>i</b> d <b>e</b> o            | <b>s</b> t <b>u</b> v <b>i</b> ew             |
| oooo 27 | <b>s</b> t <b>u</b> v <b>i</b> ll <b>a</b> g <b>e</b> | <b>s</b> t <b>u</b> v <b>i</b> n <b>t</b> a <b>g</b> e | <b>s</b> t <b>u</b> v <b>i</b> ol <b>i</b> n         | <b>s</b> t <b>u</b> v <b>i</b> r <b>tu</b> al | <b>s</b> t <b>u</b> v <b>i</b> r <b>u</b> s  | <b>s</b> t <b>u</b> v <b>i</b> s <b>a</b>            | <b>s</b> t <b>u</b> v <b>i</b> s <b>i</b> t            | <b>s</b> t <b>u</b> v <b>i</b> s <b>u</b> al | <b>s</b> t <b>u</b> v <b>i</b> t <b>a</b> l  | <b>s</b> t <b>u</b> v <b>i</b> v <b>i</b> d                     | <b>s</b> t <b>u</b> v <b>o</b> c <b>a</b> l  | <b>s</b> t <b>u</b> v <b>o</b> i <b>c</b> e   | <b>s</b> t <b>u</b> v <b>o</b> i <b>d</b>    | <b>s</b> t <b>u</b> v <b>o</b> l <b>c</b> ano         | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> m <b>e</b>   | <b>s</b> t <b>u</b> v <b>o</b> t <b>e</b>     |
| oooo 28 | <b>s</b> t <b>u</b> v <b>o</b> y <b>a</b> g <b>e</b>  | <b>s</b> t <b>u</b> v <b>o</b> g <b>e</b>              | <b>s</b> t <b>u</b> v <b>o</b> g <b>on</b>           | <b>s</b> t <b>u</b> v <b>o</b> i <b>t</b>     | <b>s</b> t <b>u</b> v <b>o</b> l <b>k</b>    | <b>s</b> t <b>u</b> v <b>o</b> ll                    | <b>s</b> t <b>u</b> v <b>o</b> l <b>n</b> ut           | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt                    | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt  | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt          | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt           | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> nt  |
| oooo 29 | <b>s</b> t <b>u</b> v <b>o</b> y                      | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th           | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th         | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th  | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th         | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th           | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th                    | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th  | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th          | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th           | <b>s</b> t <b>u</b> v <b>o</b> l <b>u</b> th  |
| oooo 30 | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                                | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l              |
| oooo 31 | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                                | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l              |
| oooo 32 | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                     | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                                | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l              | <b>s</b> t <b>u</b> v <b>o</b> l             | <b>s</b> t <b>u</b> v <b>o</b> l                      | <b>s</b> t <b>u</b> v <b>o</b> l                       | <b>s</b> t <b>u</b> v <b>o</b> l              |